Marçal Herraiz Bayó

Paris, France

Personal website | LinkedIn | GitHub

marcalhb@gmail.com | +34 662 048 927

Education

MSc in Mathematics – PSL University — Paris, France
Rank: 5/116 (top 5%); GPA: 15.052/20; Minor in Physics — Sept. 2025 – Present
- Coursework at Dauphine and École Normale Supérieure. Full FSMP scholarship.
- First-year thesis: "Market Making and Optimal Inventory Management" (18/20).

BSc in Mathematics – University of Barcelona — Barcelona, Spain
Rank: 5/141 (top 4%); GPA: 9.02/10 — Sept. 2020 – June 2025
- Thesis: "The Riemann–Hilbert Correspondence for Flat Connections on Principal Bundles" (10/10).

BSc in Physics – University of Barcelona — Barcelona, Spain
Rank: 16/196 (top 8%); GPA: 8.79/10 — Sept. 2020 – June 2025
- Thesis: "Improving Classical Shadows with Grouping Strategies" (9.7/10).

Industry Experience

Westinghouse Nuclear — Madrid, Spain
Simulation Modeling Engineer – Nuclear Plant Simulation — July 2025 – Present
- Developing a compressible liquid–vapor–air model in Fortran with revised state variables and pressure solver.
- Secured leadership approval for a new multiphase-modeling project by presenting preliminary results.
- Built and validated a condenser heat-transfer model with under 2% error against plant data.
- Resolved major energy-conservation errors in open tanks by implementing a new overflow model.
- Delivered thermal-hydraulic models to production software for projects and plants under construction.

Research Experience

Barcelona Supercomputing Center (BSC) — Barcelona, Spain
Visiting Student – Quantum Computing — Nov. 2024 – June 2025
- Implemented a hybrid algorithm combining classical shadows with observable grouping using PennyLane.
- Improved energy-estimation accuracy up to 18-fold over standard classical shadows.
- Reduced measurement requirements by a factor of 100–1,000 compared with standard classical shadows.
  Supervisors: Prof. Dr. Bruno Julià, Dr. Berta Casas, Dr. Sergi Masot.

Institute for Research in Biomedicine (IRB) — Barcelona, Spain
Research Intern – Computational Biophysics — July 2024 – Feb. 2025
- Modeled DNA flexibility using coarse-grained computational methods in C.
- Extended the model to analyze how DNA damage affects flexibility.
- Developed Python algorithms to compute DNA geometric descriptors.
  Supervisor: Prof. Dr. Modesto Orozco.

Center for Mathematical Research (CRM) — Barcelona, Spain
Research Intern – Complex Dynamics — June – Aug. 2023
- Studied bifurcations in the complex exponential family and the non-structural stability of $\exp(z)$.
  Supervisors: Prof. Dr. Núria Fagella, Prof. Dr. Xavier Jarque.

Honors and Awards

Fondation Sciences Mathématiques de Paris: full master's scholarship; 1,000+ applicants. 2025
Collaboration Scholarship: research toward the Mathematics BSc thesis. 2024
Maths4Life Award: one of five IRB grants for research in mathematics and biomedicine. 2024
CRM Award: Bachelor's thesis research with a CRM-affiliated supervisor (declined). 2024
Excellent with Honors: awarded in 16 subjects across both BSc programs at UB. 2021–2025
Honors in Baccalaureate: academic distinction covering first-year university tuition in full. 2020
Mad for Science Programme: selected for a Physics course introducing students to research. 2019

Skills and Languages

Programming Python, C, Fortran, LaTeX (proficient); R, Bash, Mathematica, SageMath, HTML, CSS (familiar).
Libraries NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, PennyLane.
Tools Git/GitHub, Unix/Linux, Jupyter.
Languages Spanish and Catalan (native) • English (C1) • French (A2).